

HSIANG YU (ANNA) HUANG

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EXPERIENCE

AI Engineer Intern

Feb. 2026 – Present

Finz (FinTech Startup)

Boston, MA (Remote)

- Built event-driven ingestion on **FastAPI**, **Kafka**, **Faust**, and **MongoDB** with idempotent upserts, plus **Playwright** connectors that pull ADP and Shift4 revenue and payroll data (credentials in AWS Secrets Manager).
- Built a vendor normalization and Chart of Accounts categorization pipeline with a two-stage fuzzy-matching classifier and a data-quality agent, reaching **98.8% auto-classification** on a client's live ledger.
- Built a **Gemini vision** pipeline that parses bills and invoices into line items written back to **QuickBooks Online**, triggered automatically by incoming email; tested successfully on 30+ bills.
- Built a production agentic CFO assistant on **Google ADK**, **Gemini (Vertex AI)**, and **LangGraph**, routing across 21 finance tools with grounded tool retrieval so answers trace back to real ledger data.

Research Assistant, LLM Platform

Sep. 2025 – Jan. 2026

BU BIT Lab

Boston, MA

- Architected a multi-agent RAG platform for generative engine optimization research on **FastAPI** and **LangGraph**.
- Designed **MongoDB** schemas for conversation history, retrieval contexts, and per-step latency/token/routing logs.

Full Stack Developer

Sep. 2024 – May 2025

Citale (BU Spark! Launch Lab)

Boston, MA

- Designed and indexed the **PostgreSQL** schema and implemented 10+ features (messaging, notifications, user profiles) with Next.js and Supabase, deployed on Vercel.

Research Assistant

Apr. 2023 – Jul. 2024

NTUST – AI & Decision Analysis Lab; Decision Analysis & Applied Statistics Lab

Taipei, Taiwan

- Built an **ARIMA-XGBoost** sales forecasting pipeline for a graphic card manufacturer, boosting R^2 from 0.083 to 0.734, and presented the model and results to a business manager.
- Clustered vending machine product sales with **K-means** and built a classification tree by price segment to recommend on-shelf products for specific locations.

PROJECTS

NVIDIA Sentiment Data & ML Pipeline | Azure Data Factory, Synapse Spark, MLlib, Power BI

- Rebuilt a BU Big Data Engineering team course project solo, end-to-end on Azure: **Data Factory** and **Azure Functions** ingest X and Yahoo Finance data into **ADLS Gen2** medallion layers as typed, deduplicated Parquet.
- Built **PySpark** feature engineering (VADER sentiment, engagement) and a **Spark MLlib** Random Forest predicting NVDA next-session volume, splitting train/test by trading session to prevent label leakage.
- Orchestrated daily prediction, post-close backfill, conditional retraining, and refresh of a **Synapse SQL** dimensional warehouse (2 dimensions, 8 fact tables) for **Power BI**, auto-pausing the SQL pool to cut idle compute.

TECHNICAL SKILLS

Languages	Python (Pandas, PySpark), SQL, TypeScript, C++ , Dart (Flutter)
Data Pipelines	Kafka, Faust, Spark, Azure Data Factory, Azure Functions, Medallion Architecture, Idempotent Ingestion
Data Stores	PostgreSQL, MongoDB, Redis, Azure Synapse (Spark, Dedicated SQL Pool), ADLS Gen2, Amazon S3
Cloud / DevOps	AWS (EC2, S3, Lambda, IAM, Secrets Manager), Azure, GCP, Docker, GitHub Actions (CI/CD), Linux
AI / LLM	LangGraph, Google ADK, Gemini (Vertex AI, Vision), RAG, Tool Calling, LLM Evals, Spark MLlib

EDUCATION

Boston University

Jan. 2026

M.S. in Data Science | GPA: 3.6/4.0 | Best Overall, BU DS+X Hackathon 2025 (RhettSearch)

Boston, MA

Relevant Coursework: Databases (C++), Artificial Intelligence, Deep Learning, Big Data Engineering

National Taiwan University of Science and Technology

Jun. 2023

B.B.A. in Industrial Management & Bachelor of Finance (Double Degree), Minor in CS | GPA: 3.85/4.3

Taipei, Taiwan

Relevant Coursework: Machine Learning, Algorithms, Object-Oriented Programming